CUME

Inorganic Chemistry

Sept-30-2011

This week's seminar speaker discussed his novel NRM and XRD studies on Li-compounds such as n-Bu-Li. The Li-chemistry in general has important applications in the development of new drug molecules, rechargeable Li-ion batteries, etc.

Question-1. Since the natural abundance of Li is low and the Li-price is high, the worldwide trend of going for rechargeable Li-ion battery may soon hit a wall. Currently, many are trying to find a substitute for the Li (e.g. a compound, a different element, etc.) with which people can develop a new rechargeable battery higher in charge-density and/or cheaper than the current Li-versions. Please find one such substitute of your choice, and explain why you believe that this substitute may work. (20')

Question-2. Compare the XRD sensitivity of the Li in a solid with that of the substitute and briefly why so. (10')

Question-3. Compare Li-atom, Li^+ and $Li^+_{(aq)}$ with those of the substitute in the HSAB theory. (15')

Question-4. Compare the first ionization energy and melting point of Li-atom with that of the substitute. (10')

Question-5. Besides NMR and XRD, what else may you use to effectively detect the Li in a compound or a solid? (5')

Question-6. Draw the structure and assign a point group for the n-Bu-Li. (10')

Question-7. Compare diameters of Li-atom, Li^+ and $Li^+_{(aq)}$ with that of Na-atom, Na^+ and $Na^+_{(aq)}$. (15')